
Beyond Replay RMSE: Auditing Offline Selection of Learned State Observers for Closed-Loop Robot Control

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Learned state observers are usually trained and selected using estimation error on
2 fixed trajectories, even though deployment places them inside a feedback loop.
3 We study the resulting proxy mismatch for differential-drive tracking with biased
4 odometry and intermittent range-bearing sensing. Six deployment candidates—
5 dead reckoning, an extended Kalman filter (EKF), and GRU or selective state-
6 space residual observers anchored to either model—are evaluated through paired
7 common-input replay and closed-loop rollouts over 24 degradation conditions.
8 Replay position RMSE preserves broad ordering (Spearman $\rho = 0.923$) but selects
9 a suboptimal closed-loop observer in 5/24 conditions. We formulate observer se-
10 lection through top-1 agreement and closed-loop regret, then test three task-aware
11 offline alternatives: exact controller-command disagreement, a local controller-
12 sensitivity error, and counterfactual error replay. The sensitivity score reduces
13 observed failures to 3/24, but its bootstrap interval overlaps no improvement and
14 its worst regret is larger; the other proxies perform worse. Matched EKF-anchor
15 ablations show that selectivity is regime-dependent rather than universally stabiliz-
16 ing, while three training initializations reveal substantial stress-regime variability.
17 A second controller changes the oracle observer in 8/24 conditions. Finally, 600
18 seconds of UTIAS MRCLAM physical logs expose poor zero-shot map transfer
19 from the ordered landmark-slot representation. We release a reproducible evalua-
20 tion protocol that treats observer deployment as a decision problem: report replay
21 and closed-loop metrics, top-1 agreement, selection regret, matched ablations,
22 training-seed uncertainty, controller dependence, and physical-log stress tests.

23 1 Introduction

24 State estimation is usually evaluated as an isolated prediction problem. A filter receives controls
25 and measurements, produces a pose, and is ranked by position or heading root-mean-square error
26 (RMSE) against ground truth. In an autonomous system, however, the estimate is consumed by a
27 controller. The resulting command changes the future state, which changes the next measurement
28 and therefore the future estimator input. This feedback creates two gaps between replay accuracy and
29 deployment utility: errors are directionally weighted by the controller, and the estimator induces its
30 own closed-loop data distribution.

31 This distinction is especially important for learned observers. Recurrent filters can exploit temporal
32 patterns and model mismatch, but they can also learn corrections that are benign on expert-generated
33 logs and destabilizing when their own errors alter the trajectory. Closely related objective mismatch
34 has been documented for learned dynamics models in model-based reinforcement learning (Lambert
35 et al., 2020); here we isolate it at the observer layer. The question is not whether replay RMSE is

36 correlated with control—it generally is—but whether it reliably supports the actual model-selection
37 decision.

38 We study a controlled yet nontrivial setting: a differential-drive robot follows a figure-eight using
39 biased, noisy wheel odometry and intermittent range-bearing measurements to a known landmark
40 map. The candidate set spans classical and hybrid estimators: dead reckoning, an EKF (Thrun et al.,
41 2005), and recurrent residual observers with GRU or selective diagonal state-space cores inspired by
42 S4 and Mamba (Gu et al., 2022; Gu and Dao, 2024). Residuals are anchored either to dead reckoning
43 or to the EKF, allowing us to separate recurrent architecture from analytic prior strength.

44 Our contribution is an evaluation framework rather than a claim that one recurrent cell dominates.
45 Specifically, we provide:

- 46 • a paired replay/closed-loop benchmark with six observers, four physical degradation axes, 24
47 conditions, and ten evaluation seeds;
- 48 • a decision-theoretic analysis using top-1 agreement, absolute regret, and relative regret, rather
49 than correlation alone;
- 50 • three controller-aware offline proxy baselines and an empirical demonstration that task awareness
51 does not automatically yield deployment-valid selection;
- 52 • matched same-anchor ablations, a three-training-seed robustness study, and a matched pure-
53 pursuit/Kanayama controller sensitivity analysis;
- 54 • physical-log localization and deliberately difficult zero-shot transfer on UTIAS MRCLAM (Leung
55 et al., 2011), plus a real-noise-calibrated closed-loop replay.

56 The central finding is nuanced: replay position RMSE is an informative coarse proxy, but it does not
57 eliminate the need for a small, stratified closed-loop evaluation.

58 2 Related work

59 **Classical and learned state estimation.** The EKF remains a standard nonlinear localization base-
60 line (Thrun et al., 2005). Learned estimators range from discriminative recurrent filters (Haarnoja
61 et al., 2016) and variational state-space models (Karl et al., 2017; Rubanova et al., 2019) to archi-
62 tectures that preserve algorithmic structure, including recurrent Kalman networks (Becker et al.,
63 2019), differentiable particle filters (Jonschkowski et al., 2018), and KalmanNet (Revach et al., 2022).
64 Our residual observers follow this hybrid philosophy but focus on how candidates are selected for
65 feedback deployment.

66 **State-space sequence models.** Structured state-space models provide efficient long-context recur-
67 rences (Gu et al., 2022); selective SSMs make transition and input parameters depend on the current
68 token (Gu and Dao, 2024). Intermittent sensing appears naturally suited to selectivity because the
69 observer should adapt retention and correction when measurements vanish or reappear. We test that
70 intuition with a compact diagonal selective cell and, importantly, with matched non-selective and
71 mask-free controls.

72 **Task-aware evaluation and proxy mismatch.** Pure pursuit (Coulter, 1992) and Kanayama track-
73 ing (Kanayama et al., 1990) map pose errors into actions differently, so an estimator’s utility can
74 be controller-specific. Proxy mismatch in model-based RL (Lambert et al., 2020) motivates our
75 emphasis on selection regret. Unlike work proposing a single task-aware loss, we compare several
76 offline surrogates and report their failure modes.

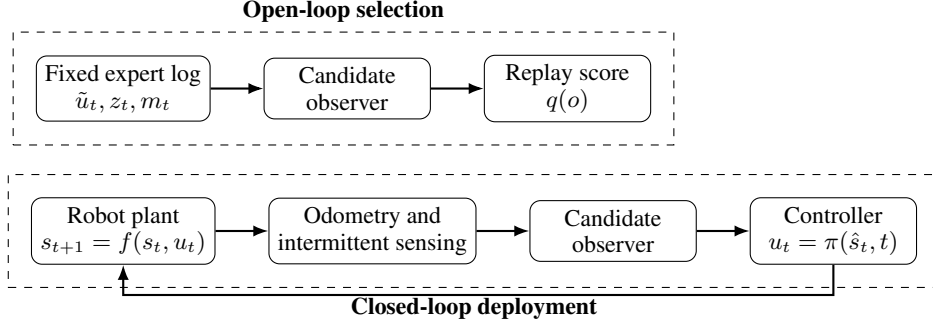


Figure 1: Replay holds inputs fixed, while closed-loop evaluation lets observer errors alter actions, states, and future measurements.

3 Closed-loop observer selection

3.1 Robot, sensing, and feedback

The robot state is $s_t = (x_t, y_t, \theta_t)$ and the commanded input is $u_t = (v_t, \omega_t)$. With step size $\Delta t = 0.1$ s,

$$\begin{aligned} x_{t+1} &= x_t + \Delta t v_t \cos \theta_t, \\ y_{t+1} &= y_t + \Delta t v_t \sin \theta_t, \\ \theta_{t+1} &= \text{wrap}(\theta_t + \Delta t \omega_t). \end{aligned} \quad (1)$$

Odometry contains multiplicative linear-velocity slip, additive angular-rate bias, and Gaussian noise: $\tilde{v}_t = \kappa v_t + \epsilon_t^v$ and $\tilde{\omega}_t = \omega_t + b_g + \epsilon_t^\omega$. Visible landmark j yields noisy range and bearing $z_{t,j} = (\|p_j - p_t\| + \epsilon^r, \text{atan2}(p_j - p_t) - \theta_t + \epsilon^b)$. A periodic schedule removes all exteroceptive measurements during blackouts; a mask indicates which landmark slots are present.

The controller is $u_t = \pi(\hat{s}_t, t)$, where $\hat{s}_t$ is the observer output. Thus the closed-loop state distribution depends on the candidate observer. Figure 1 contrasts common-input replay with the deployment loop.

3.2 Observer selection as a decision problem

Let $\mathcal{O}$ be the candidate set, c an operating condition, $q_c(o)$ an offline score, and $J_c(o)$ the closed-loop cross-track RMSE. The proxy-selected and oracle observers are

$$o_{q,c}^* = \arg \min_{o \in \mathcal{O}} q_c(o), \quad o_{\text{CL},c}^* = \arg \min_{o \in \mathcal{O}} J_c(o). \quad (2)$$

We report top-1 failure $\mathbb{1}[o_{q,c}^* \neq o_{\text{CL},c}^*]$, absolute regret $\mathcal{R}_{q,c} = J_c(o_{q,c}^*) - J_c(o_{\text{CL},c}^*)$, and relative regret $\mathcal{R}_{q,c}/J_c(o_{\text{CL},c}^*)$. This explicitly evaluates the deployment decision; a high global correlation can coexist with repeated top-1 errors.

4 Observer models and offline proxies

4.1 Analytic and residual observers

Dead reckoning integrates $\tilde{u}_t$. The EKF uses the same unicycle motion model and masked range-bearing updates. A learned observer first advances an analytic anchor $\bar{s}_t$ and then predicts a residual:

$$h_t = \mathcal{G}_\psi(h_{t-1}, \phi_t), \quad \hat{s}_t = \bar{s}_t \oplus W_o h_t. \quad (3)$$

The feature ϕ_t contains odometry increments; eight ordered tuples $(r_j, \sin b_j, \cos b_j, m_j)$; a global measurement-presence bit; anchor position and sinusoidal heading; and, for EKF-anchored models, the covariance diagonal. The near-zero readout initialization makes the initial model approximately equal to its anchor.

Table 1: Principal observer set. All learned models use hidden size 32 and the same residual loss.

Observer	Anchor	Recurrent core	Covariance input	Parameters
DR	Dead reckoning	–	–	–
EKF	EKF	–	internal	–
GRU-DR	Dead reckoning	GRU	no	7,011
SSM-DR	Dead reckoning	selective SSM	no	5,219
GRU-EKF	EKF	GRU	yes	7,299
SSM-EKF	EKF	selective SSM	yes	5,603

102 For the selective diagonal SSM, with $A = -\text{softplus}(a)$,

$$\begin{aligned}
\Delta_t &= \text{softplus}(W_\Delta \phi_t + b_\Delta), \quad B_t = W_B \phi_t + b_B, \quad C_t = W_C \phi_t + b_C, \\
\bar{A}_t &= \exp(\Delta_t \odot A), \\
h_t &= \bar{A}_t \odot h_{t-1} + (\Delta_t \odot B_t) \odot (W_{\text{in}} \phi_t), \\
r_t &= W_o(C_t \odot h_t) + b_o.
\end{aligned} \tag{4}$$

103 The non-selective ablation replaces Δ_t, B_t, C_t by learned constants. This is a compact selective
104 recurrence, not a full Mamba block.

105 4.2 Controller-aware offline proxies

106 Position and heading RMSE ignore the downstream map from pose to action. We additionally test
107 three log-only proxies. Let $e_t = \hat{s}_t \ominus s_t$, D normalize action units, and $J_\pi(s_t, t)$ be a finite-difference
108 controller Jacobian.

$$q_{\text{cmd}}(o) = \left[\frac{1}{T} \sum_t \|D^{-1}(\pi(\hat{s}_t, t) - \pi(s_t, t))\|_2^2 \right]^{1/2}, \tag{5}$$

$$q_{\text{LCSE}}(o) = \left[\frac{1}{T} \sum_t \|D^{-1} J_\pi(s_t, t) e_t\|_2^2 \right]^{1/2}. \tag{6}$$

109 Counterfactual error replay injects the logged error sequence e_t into a shadow plant and evaluates
110 the tracking error produced by the controller. It captures temporal accumulation but still freezes the
111 observer error trace, so it does not reproduce full feedback adaptation.

112 5 Experimental design

113 **Training.** Each recurrent observer is trained on 200 expert-controlled trajectories of 200 steps, with
114 a separate validation set. Every trajectory randomizes path amplitude, slip, gyro bias, range/bearing
115 noise, blackout length and period, and active landmark count. We use hidden size 32, Adam, gradient
116 clipping at 5, batch size 24, 32 epochs, and retain the best validation checkpoint. The loss is squared
117 position error plus $0.5(1 - \cos \Delta\theta)$.

118 **Synthetic evaluation.** The reference contains 220 steps. At each of 24 conditions, the six observers
119 are evaluated over ten fresh seeds in both common-input replay and closed loop. One degradation
120 variable is swept while all others remain nominal. Table 2 lists the full design. Closed-loop metrics
121 include cross-track and pose RMSE, maximum cross-track error, angular-control effort, post-blackout
122 recovery time, and divergence fraction.

123 **Physical logs.** UTIAS MRCLAM contains synchronized odometry, range–bearing observations,
124 known landmark positions, and Vicon ground truth (Leung et al., 2011). We align a 600 s segment of
125 Dataset 1 Robot 1 to 6,000 samples with 1,878 observations. We evaluate dead reckoning, a 3×3
126 EKF covariance grid, and zero-shot learned models on an eight-landmark subset. A separate synthetic
127 replay resamples empirical sensor residuals and the physical observation-count sequence; it is not
128 presented as hardware closed-loop control.

Table 2: Evaluation protocol. Range-noise sweeps use bearing noise $\sigma_b = 0.5\sigma_r$.

Component	Values
Nominal sensing	slip 0.92; gyro bias 0.06 rad/s; $\sigma_r = 0.12$ m; $\sigma_b = 0.06$ rad; blackout 14/45 steps; 6 landmarks
Blackout sweep	4, 10, 16, 22, 28, 34 steps
Range-noise sweep	$\sigma_r \in \{0.05, 0.12, 0.20, 0.30, 0.40, 0.50\}$ m
Landmark sweep	2, 3, 4, 5, 6, 8 active landmarks
Gyro-bias sweep	0, 0.05, 0.10, 0.15, 0.20, 0.25 rad/s
Primary controller	pure pursuit, $v_{\text{cmd}} = 1.0$ m/s, lookahead 0.8 m
Secondary controller	Kanayama tracking, matched plant, limits, checkpoints, and five seeds
Uncertainty	95% CI across evaluation seeds; condition bootstrap for selection; three SSM-EKF training seeds

Table 3: Nominal evaluation over ten seeds, mean $\pm$ 95% CI half-width. Lower is better. “Div.” is the fraction of steps outside the 1 m tracking tube.

Observer	Replay pos. (m)	Replay head. (rad)	CL cross-track (m)	CL pose (m)	Max cross-track (m)	Effort $\overline{\omega^2}$	Div.
DR	1.503 ± 0.200	0.832 ± 0.022	1.045 ± 0.025	1.834 ± 0.040	2.557 ± 0.062	0.449 ± 0.063	0.260
EKF	0.140 ± 0.009	0.092 ± 0.006	0.212 ± 0.018	0.146 ± 0.006	0.628 ± 0.153	0.681 ± 0.128	0.004
GRU-DR	0.281 ± 0.031	0.359 ± 0.087	0.694 ± 0.345	1.033 ± 0.371	1.718 ± 0.781	0.914 ± 0.129	0.140
SSM-DR	0.325 ± 0.039	0.276 ± 0.038	0.909 ± 0.498	1.399 ± 0.809	2.016 ± 0.947	1.123 ± 0.120	0.195
GRU-EKF	0.094 ± 0.012	0.093 ± 0.015	0.163 ± 0.022	0.109 ± 0.013	0.573 ± 0.085	0.949 ± 0.128	0.000
SSM-EKF	0.121 ± 0.018	0.090 ± 0.013	0.199 ± 0.033	0.191 ± 0.043	0.684 ± 0.076	1.052 ± 0.136	0.000

129 **Statistics.** We compute global Spearman/Pearson associations over 24×6 condition-observer pairs,
130 per-condition rank agreement, top-1 failures, and selection regret. Confidence intervals for proxy
131 selection resample the 24 conditions. Matched ablations use paired evaluation seeds. The controller
132 comparison bootstraps the five matched seeds and is reported as a sensitivity analysis.

133 6 Results

134 6.1 Nominal performance and degradation regimes

135 Table 3 shows the nominal operating point. EKF-anchored learned observers are the only learned
136 models that remain consistently stable. GRU-EKF attains the best replay and closed-loop performance,
137 reducing cross-track RMSE from 0.212 m for EKF to 0.163 m. In contrast, GRU-DR and SSM-DR
138 improve replay RMSE by factors of 4–5 relative to dead reckoning while retaining much larger
139 closed-loop error and divergence. This is an immediate example of an improvement that does not
140 survive feedback.

141 Figure 2 reveals that the best closed-loop observer is regime-dependent. GRU-EKF dominates
142 moderate blackouts and most noise/bias levels. Plain EKF becomes preferable at the two longest
143 blackouts, where learned correction has little information to exploit. SSM-EKF is strongest with two
144 active landmarks and is competitive under high range noise and bias. DR-anchored recurrent models
145 degrade sharply as blackout duration and bias grow.

146 6.2 Replay accuracy is correlated with control, but not decision-complete

147 Across all 144 condition-observer pairs, replay position RMSE has Spearman $\rho = 0.923$ and Pearson
148 $r = 0.477$ with closed-loop cross-track error (Figure 3). The high rank correlation means replay
149 evaluation is useful for coarse screening. The weaker magnitude correlation and top-1 failures mean
150 it is insufficient for final deployment selection.

151 Replay position RMSE selects a different observer than the closed-loop oracle in 5/24 conditions:
152 the two longest blackouts, the two highest range-noise levels, and the eight-landmark endpoint. Mean
153 regret is 0.0051 m and maximum regret is 0.0347 m (12.7% relative). These numbers are modest
154 in this low-speed simulation, but the pattern is systematic: replay favors SSM-EKF when EKF or
155 GRU-EKF has lower tracking error.

156 LCSE reduces the observed failure count from five to three by accounting for controller anisotropy, but
157 the paired bootstrap interval for failure-fraction reduction is $[-0.125, 0.292]$. Its two-landmark failure

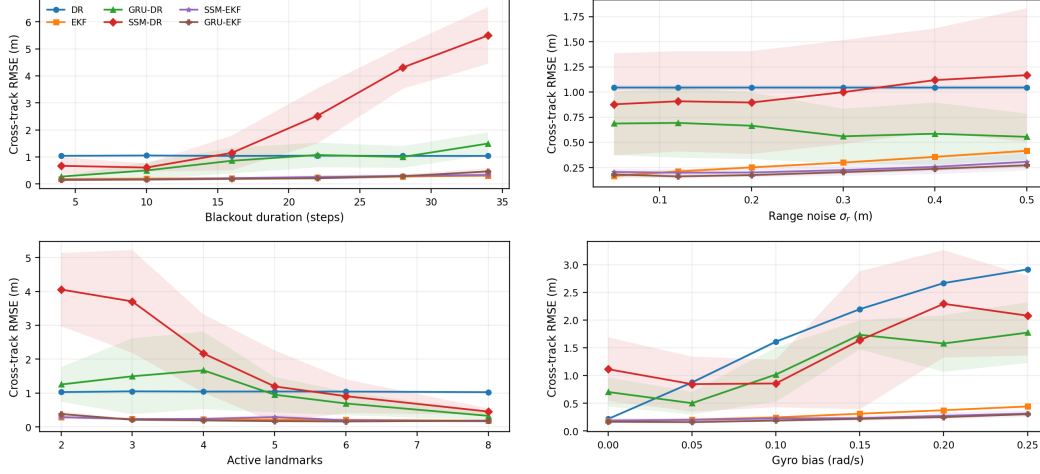


Figure 2: Corrected six-observer closed-loop sweeps, mean $\pm$ 95% CI over ten evaluation seeds. No single learned architecture is uniformly best.

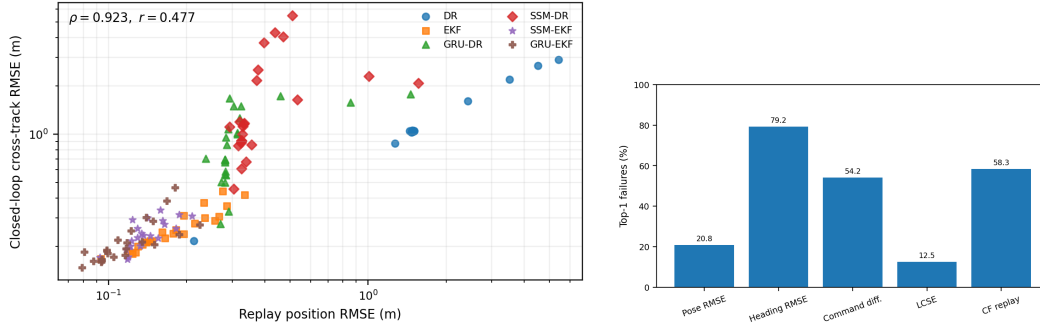


Figure 3: Left: replay position RMSE versus closed-loop tracking over all 144 condition-observer pairs. Right: top-1 deployment failures for five offline proxies.

incurs 0.094 m regret, larger than the worst replay-RMSE failure. Exact command disagreement is worse because action similarity at a fixed state does not capture future state and sensing changes; counterfactual replay likewise freezes the observer error trace. Controller awareness therefore shifts rather than eliminates failure regimes.

Compound stress. When all axes are simultaneously hard, GRU-EKF is best in replay and closed loop (Table 5). The sharper counterexample is within SSM-EKF: relative to EKF it reduces replay position RMSE by 43% ($0.829 \rightarrow 0.472$ m) yet increases closed-loop cross-track RMSE by 10% ($1.324 \rightarrow 1.459$ m). Therefore, a large real replay improvement for a particular observer need not transfer to control.

6.3 Matched ablations: the anchor is more consistent than selectivity

A valid selectivity ablation must keep the analytic anchor, inputs, hidden size, and training protocol fixed. Table 6 compares three EKF-anchored SSMs. Nominal performance is nearly identical. Selectivity helps under high gyro bias but hurts under long blackouts and sparse sensing. Removing the global mask bit has little effect because each landmark tuple already contains a local mask. The robust qualitative factor is the EKF anchor: DR-anchored learned models are much less stable despite competitive replay error.

6.4 Training uncertainty and controller dependence

The three SSM-EKF initializations have nominal means from 0.189 to 0.199 m, but long-blackout means from 0.231 to 0.336 m and high-noise means from 0.249 to 0.307 m. Evaluation-seed

Table 4: Offline proxy quality against closed-loop cross-track RMSE. Failure and regret intervals bootstrap the 24 conditions.

Proxy	ρ	Failures	Failure CI	Mean regret	Max
Position RMSE	.923	5/24	[.083,.375]	.0051	.0347
Heading RMSE	.881	19/24	[.625,.958]	.0237	.0474
Command disagreement	.876	13/24	[.333,.750]	.0188	.1208
LCSE	.903	3/24	[.000,.250]	.0051	.0940
Counterfactual replay	.842	14/24	[.375,.792]	.0159	.0474

Table 5: Compound-stress evaluation, mean $\pm$ 95% CI.

Observer	Replay pose (m)	CL pose (m)	CL cross-track (m)
DR	3.289 $\pm$ 0.382	4.818 $\pm$ 0.030	2.528 $\pm$ 0.014
EKF	0.829 $\pm$ 0.066	1.467 $\pm$ 0.358	1.324 $\pm$ 0.094
GRU-EKF	0.447 $\pm$ 0.049	1.222 $\pm$ 0.422	1.052 $\pm$ 0.326
SSM-EKF	0.472 $\pm$ 0.044	1.496 $\pm$ 0.350	1.459 $\pm$ 0.156
SSM-DR	0.798 $\pm$ 0.099	5.734 $\pm$ 1.419	4.228 $\pm$ 1.360

confidence intervals around one checkpoint therefore understate uncertainty about architecture-level stress robustness.

Controller choice also changes the deployment decision. In a matched five-seed sensitivity analysis, replacing pure pursuit with Kanayama tracking changes the closed-loop oracle in 8/24 conditions (33.3%; paired bootstrap CI 20.8–62.5%). Replay RMSE misselects the Kanayama observer in 7/24 conditions, with mean regret 0.0142 m and maximum regret 0.0923 m. These results are lower-powered than the primary ten-seed study, but they show that “best observer” is partly a property of the observer-controller pair.

6.5 Physical logs expose map-transfer limitations

On the 15-landmark MRCLAM log, dead reckoning reaches 3.392 m position RMSE. The best EKF covariance setting reaches 1.357 m and improves heading RMSE from 1.662 to 0.850 rad (Figure 4). This confirms that the physical data contain useful landmark corrections and provides a real-sensor sanity check.

The learned input representation, however, has exactly eight ordered landmark slots and was trained on one synthetic map. We therefore evaluate it without retraining on an eight-landmark MRCLAM subset. GRU-EKF and SSM-EKF modestly improve position RMSE over dead reckoning (2.597 and 2.625 versus 2.822 m) but worsen heading RMSE (1.664 and 1.650 versus 1.509 rad). This should not be interpreted as successful transfer. It directly identifies the architecture’s missing invariance: landmark observations should be encoded as a set, using permutation-invariant aggregation (Zaheer et al., 2017; Lee et al., 2019), and training should randomize map geometry and cardinality.

The empirical-noise closed-loop replay provides a further caution. In this low-bias physical segment, dead reckoning is the closed-loop oracle among the tested EKF covariance grid. Pose RMSE selects it correctly, whereas command disagreement and LCSE favor more aggressive EKFs and incur 0.120 and 0.203 m regret. Sensor realism alone does not make an offline proxy closed-loop valid when the observer and sensing process do not adapt to the counterfactual trajectory.

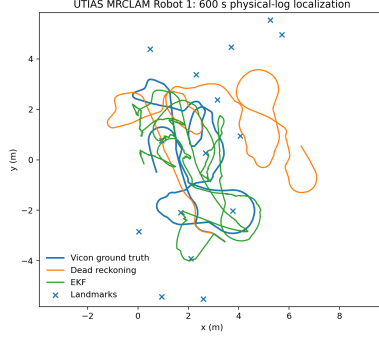
7 Discussion

Replay is useful but not decision-complete. The high global rank agreement shows that replay RMSE is a sensible coarse screen. Its failures nevertheless cluster in regimes where recurrent correction changes value: long information gaps, high measurement noise, and altered landmark availability. Controller-aware proxies recover some of these cases but create others, so the practical alternative is not another universal scalar. A small closed-loop budget should be allocated to the stress regimes in which candidate rankings are close or the sensing process changes qualitatively.

Architecture and reporting implications. The EKF anchor is the most consistent stabilizer. Selectivity helps under bias but can hurt during long blackouts and sparse sensing. MRCLAM

Table 6: Matched EKF-anchor ablation: closed-loop cross-track RMSE (m). Bold is best per condition.

Condition	Selective+mask	Non-selective	No global mask
Nominal	0.199	0.203	0.195
Long blackout	0.336	0.235	0.308
High range noise	0.307	0.322	0.294
Two landmarks	0.289	0.235	0.274
High gyro bias	0.314	0.393	0.312



Setting	Observer	Pos.	Head.
15 LM	Dead reckoning	3.392	1.662
15 LM	Best EKF	1.357	0.850
8-LM OOD	Dead reckoning	2.822	1.509
8-LM OOD	EKF	2.990	1.514
8-LM OOD	GRU-EKF	2.597	1.664
8-LM OOD	SSM-EKF	2.625	1.650

The left panel uses 600 seconds of physical sensing and Vicon ground truth. Learned models on the right are applied zero-shot to an eight-landmark subset and are not evidence of successful transfer.

Figure 4: UTIAS MRCLAM Dataset 1 Robot 1 physical-log localization and zero-shot map-transfer stress test.

exposes a separate representation failure: ordered landmark slots bind the learned model to one map. Future observers should combine analytic innovations with permutation-invariant measurement-set encoders (Zaheer et al., 2017; Lee et al., 2019), randomized maps, and calibrated uncertainty. For deployment-oriented learned filters, we recommend reporting common-input replay and matched closed-loop rollouts, rank correlation and selection regret, same-anchor ablations, separate training- and evaluation-seed uncertainty, at least one controller shift, and a clear distinction among physical logs, calibrated simulation, and hardware control.

Limitations. The plant is simulated in closed loop; data association and the map are known; the primary study uses one path family and controller; the Kanayama analysis has five seeds; only SSM-EKF is retrained across initializations; and MRCLAM covers one robot and sequence. These constraints preclude claims of hardware robustness. The highest-value next steps are randomized-map set encoders, multi-seed training for all learned observers, additional MRCLAM sequences with occlusions, controller-gain sweeps, and a ROS/Gazebo or physical closed-loop study.

8 Conclusion

Replay position RMSE preserves broad ordering but selects a different closed-loop winner in 5/24 intermittent-sensing conditions. Task-aware offline scores expose controller sensitivity without consistently solving selection. Same-anchor, training-seed, controller, and physical-log analyses show that deployment quality depends on the full observer-controller-environment system. Learned observers should therefore be selected with replay metrics plus a small, stratified closed-loop evaluation.

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